UNITED REPUBLIC OF TANZANIA
PRESIDENTS' OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
IFAKARA TOWN COUNCIL
FORM FOUR PRE-MOCK EXAMINATION

033/1 **BIOLOGY 1**

Time: 3 Hours **Monday, 24th March 2025.**

Instructions

1. This paper consists of Section A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in section A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty-four (54)** marks and section C **thirty (30)** marks.
4. Non programmable calculators may be used.
5. Communication devices, calculators and any unauthorised materials are **not** allowed in the assessment room.
6. Write your **Examination Number** on every page of your answer booklet(s)

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SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) -(x), choose the correct answer among the given alternatives and write its letter beside the item number in the answer booklet provided

(i) Which of the following is the most important difference between plants and animals?

A Growth B Movement C Nutrition
D Sensitivity E Development

(ii) The following are the branches of biology. Identify one which is not

A Microbiology B Ecology C Genetics
D Science E Mycology

(iii) Which of the following specialized cells in plant is responsible for the high concentration of chloroplast responsible for photosynthesis?

A Chlorophyll B Palisade C Guard cell
D Root hair cells E Xylems

(iv) The following are examples of water borne diseases:

A Malaria and Bilharzias B Diarrhea and malaria
C Yellow fever and Typhoid D Cholera and Typhoid
E Malaria and Typhoid

(v) Mutual interaction between two species is described by which of the following statement?

A Both lives as parasite B Both benefit and flourish
C One harmed and the other benefit D Both neither benefit nor harmed
E Non –of the above is correct

(vi) A very young tadpole breath by means of

A Spiracle B Gills C Segments
D Nostril E Skin

(vii) Salome gets an accident and found that she could hardly maintain her body equilibrium. Which one among the following structures of her body was probably damaged?

A Cochlea B Medullar Oblongata C Semi secular canal
D pituitary canal E Auditorium canal

(viii) Pancreases secrete pancreatic juice which contain the following enzymes

A Bile, Lipase and Trypsin B Trypsin, Bile and Amylase
C Amylase, Trypsin and mucus D Renin, Trypsin and amylase
E Trypsin, mucin and lipase

(ix) -------- is the force responsible for pulling of water to the top of the tall trees

A Capillarity pull B Transpiration pull C Osmotic pressure
D Turgor pressure E Mass flow rate

(x) -------- and --------- grows into shoot and root respectively during seed germination

A Plumule and Epicotyl B Plumule and radicle
C Radicle and Plumule D Seed and Hilum
E Epicotyl and plumule

2. Match the responses in List A with responses in List B by writing the letters of the correct response beside the item number in the answer booklet provided.

LIST A	LIST B
(i) Maintenance of constant internal environment	A. Diabetes mellitus
(ii) Deamination of Amino Acid	B. Nephron
(iii) Blood capillaries within Bowman's capsule	C. Kidney
(iv) Sugar in urine	D. Glomerulus
(v) Functional unit of a kidney	E. Liver
(vi) Dilute urine	F. Diabetes inspindus
	G. Regulation
	H. Homeostasis
	I. Urination
	J. Excretion

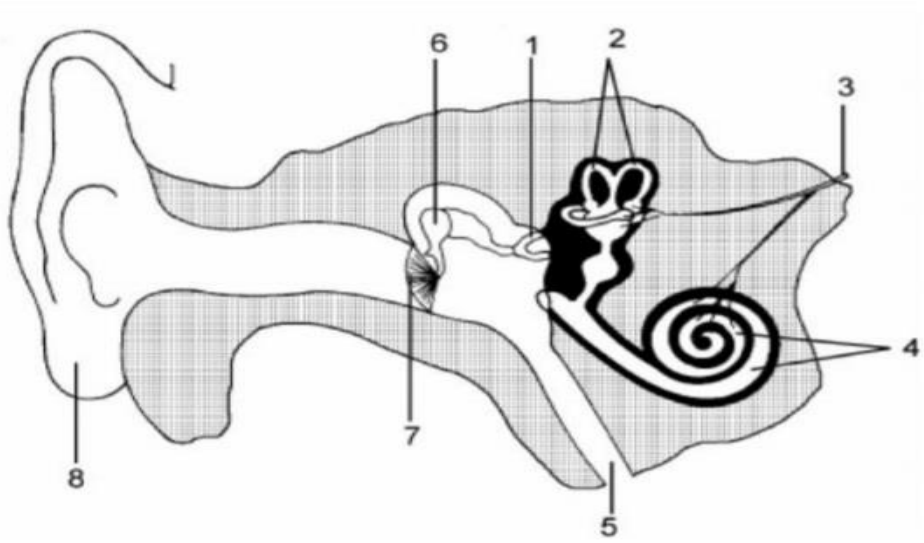
SECTION B (54 Marks)

Answer **all** questions in this section

3. (a). Name the tissues in plants responsible for:
(i) Transport of water and mineral salts
(ii) Transport of carbohydrates
(iii) Primary growth
(b). State five ways how the ileum is adapted for absorption function.
(c). Why is movement necessary in animals? Give five reasons

4. (a). Give three reasons to why starch and not glucose is a good food storage in plants
(b). Explain why it is: -
(i) Not health to sleep in a closed room with many potted plants
(ii) Dangerous to sleep in a poorly ventilated room with a charcoal burner on.
(iii) Animal cell bursts when placed in pure water whereas plant cell does not.
(iv) Omnivores are regarded as opportunistic feeders?

5. Study the diagram below then answer the following questions



i. Name the part from 1 to 8
ii. What is the function of the part labelled 2 and 3?
iii. Why it is harmful to remove the wax from the ear using the sharp object?

6. (a) Motor vehicles move, use energy and produce carbon dioxide and water. Similar Characteristics Occur in living organisms yet motor vehicles are not classified as living. Explain why?
(b) Explain four reasons why it is healthier to breathe through nose than mouth.
(c) Why is oxygen important in the process of active transport in cells?
(d) Why are green plants referred to as primary producers in an ecosystem?
(e) Explain why images that form on the blind spot are not perceived?

7. Study the table below, then answer the questions that follows: -

S/N	Common name	Scientific name
a	Common frog	*rana Temporaria*
b	Onion	*Allium Ceppa*
c	Coconut palm	Cocos nucifera
d	Man	*Sapiens Homo*

i) In each scientific name given is incorrect, Identify the error in each case.
ii) Write the correct names in each case by considering the rules for binomial nomenclature.
iii) Why organisms belonging to different species do not interbreed to produce viable offspring?

8. (a) Explain the following:

i. Why arteries do not need valves?

ii. Why is not advisable to induce vomiting to a victim who has taken corrosive poison?

iii. Explain why viruses are considered as both living and non-living things?

(b) Name the causative agents of the following diseases: -

(i) Gonorrhoea

(ii) Syphilis

iii) Genital herpes

SECTION C (30 MARKS)

Answer only Two questions in this section.

9. Form four student after making a simple tour at Serengeti national parks saw different organisms including: - Snake, Grasshopper, Rabbit, Grass, Toad and Hawk.
   a. Construct the food web based on the organisms given above.
   b. Construct 3 food chains from the food web constructed in (a).
   c. What would happen if: -
      i. All the grass wither and die?
      ii. All the snakes are killed?
      iii. The rabbits are doubled in number?
      iv. The number of toads multiplies?
   d. From the food web constructed identify the, primary consumers and secondary consumers.

10. (a) What are the five major differences between Deoxyribonucleic acid (DNA) Ribonucleic Acid (RNA).

(b) Ngokwape has a blood group A and he married Tuju with a blood group B. After few years Gituruge was born with a blood group O. Ngokwape blamed her wife Tuju that she committed adultery with another person. Assume you are invited to resolve the conflict, by using genetic diagrams show you are going to give conclusion.

11. Write an essay on the economic importance of monera kingdom using the following guidelines

(a)Meaning of monera

(b)The advantages of monera kingdom.

(c) Disadvantages of monera kingdom.